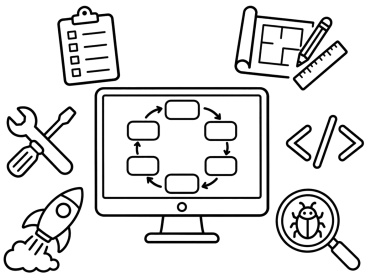
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 11 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 2
System Development Life Cycle



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 2 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 3, textbook links, a quiz
- **Answers** - chapters 1 to 3, in the same order as the workbook
- **Unit 2 Test** - the marking scheme, common mistakes and textbook notes
- **Writing spaces** - what the students write in every space

★ BEFORE THE FIRST LESSON

- ☐ Arrange a visit to the school library with the librarian (Practicals 1 and 2).
- ☐ Read the writing-space notes at the end of this guide for each lesson.
- ☐ A projector or a chart of the SDLC cycle, the waterfall and Figure 2.7.

"Show your students that every school, library and office runs on systems - and that they can build them."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Input, process or output?	8
	B Put the process steps in order	4
	C Manual or computer based?	8
	D Choose the correct answer	5
Chapter 2	A Label the life cycle	6
	B Which method?	8
	C True or false	6
	D Which level of testing?	8
	E Which deployment approach?	8
Chapter 3	A Complete the waterfall	4
	B True or false	5
	C Waterfall or iterative incremental?	8
	D Name the four models	4
	E Crossword	10
End	Unit 2 Test	40
TOTAL		**132**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Start with the two machines of Figure 2.1 and 2.2: the same parts, but only the connected ones work. That is a system.
- Ask the class for systems they know - the school, the canteen, a bus service - and name the input, the process and the output of each.
- Draw Input, Process and Output as three boxes on the board and leave them up for the whole unit.
- Bring a filled class register or a student file (with the names covered) to show a manual system.
- Data and information: in the student information system the name and the year go in, and the progress report comes out as information.
- Visit the school library before Practical 1 and arrange the visit with the librarian.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (25)	Ch 2 (36)	Ch 3 (31)	Test (40)	Total (132)
1						
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TEACHER GUIDE

Practicals at a Glance

The Activities of textbook pages 55, 56 and 63

Practical	Students do	Check that ...
1 (pp. 55, 56)	compare getting the progress report from the manual and from a computer based system; list the benefits of a computer based library system	the advantages and disadvantages follow the table of 2.6 (errors, efficiency, storage space, security); at least four library benefits of 2.5.
2 (pp. 57 - 58)	gather the requirements of a school library system by observation, an interview, a questionnaire and document samples	each method is used as in the table of 2.8; the requirements are written as needs of the librarian.
3 (p. 63)	compare and contrast the waterfall model and the iterative incremental model	waterfall: linear, one stage after the other, the user sees the system at the end; iterative incremental: one small component at a time, grows with each iteration, begins with a rough idea.

! ORGANISATION AND MATERIALS

- Practicals 1 and 3 are the textbook Activities of pages 55, 56 and 63; Practical 2 is written from pages 57 - 58 (the textbook's own library examples).
- Practicals 1 and 2 need a short visit to the school library: ask the librarian a few days before.
- Groups of four suit Practical 2: one method for each student, then they share their findings.
- No computers are needed in this unit. A projector helps to show Figures 2.5 - 2.7 and the SDLC cycle.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - Systems and information systems (2.1 - 2.6)	53 - 56	p. 55, p. 56
Chapter 2 - The system development life cycle (2.7 - 2.13)	56 - 61	-
Chapter 3 - Life cycle models (2.14 - 2.16)	61 - 63	p. 63
Unit 2 in One Page	63 (summary)	-

The textbook numbers its sections 2.1 (pages 53 - 56), 2.2 (page 56), 2.2.1 (pages 57 - 61) and 2.2.2 (pages 61 - 63). The workbook numbers its own sections 2.1 - 2.16.

★ TEACHING TIPS

- Let one group act as the manual library and one as the computer based library: who finds a book faster?
- Show a printed screenshot of a library program as an example of a prototype (page 58).

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What is a system?	**components that interact to achieve a specific task**
2. Name the three basic components of a system.	**input, process, output**
3. What does an information system do?	**converts data into information**
4. Name the two kinds of information systems.	**manual, computer based**
5. What is the first step of the SDLC?	**identification of requirements**
6. Name two methods of gathering information.	**e.g. observation, interviews**
7. In which step are the tests planned?	**designing the solution**
8. Which testing is done with the users?	**acceptance testing**
9. Old and new systems used together - which approach?	**parallel deployment**
10. Which model shows linear development?	**the waterfall model**

ANSWERS

Chapter 1

Systems and information systems

A Input, process or output?

8 marks

It is the ...	Input	Process	Output
Children (the school system)	☑	☐	☐
Providing good citizens to the country (the school system)	☐	☐	☑
The teaching/learning process with teachers and other resources	☐	☑	☐
Atmospheric pressure, temperature, wind direction and humidity	☑	☐	☐
Tomorrow's weather (the weather forecasting system)	☐	☐	☑
Name of student and relevant year (the student information system)	☑	☐	☐
Find the personal file according to the index number	☐	☑	☐
Progress report on the student for the relevant year	☐	☐	☑

1 mark each. Examples 1 and 2 of page 54 and the student information system of page 55.

B Put the process steps in order

4 marks

a = **4** b = **1** c = **3** d = **2**

1 mark each (page 55): index number, personal file, extract the information, prepare the report.

C Manual or computer based?

8 marks

The system ...	Manual	Computer based
Filing cabinets are required to store the files.	☑	☐
Security can be ensured with backups and passwords.	☐	☑
Since the data is processed by hand, there is more room for error.	☑	☐
Large amounts of data can be stored in a small physical space.	☐	☑
Data is open to a lot of threats.	☑	☐
Fines for late returns are calculated automatically.	☐	☑
All the processes are done manually.	☑	☐
Data can be processed more efficiently.	☐	☑

1 mark each. The table of page 56; fines are a benefit of the computer based library system (page 55).

D Choose the correct answer

5 marks

1. (b) input, process, output	**2.** (a) an input
3. (b) computer based	**4.** (d) more filing cabinets
5. (c) efficient processing	

★ TEACHING TIPS

- Use the table of 2.6 for a quick debate: manual systems against computer based systems.
- Draw the SDLC as a circle on the board. Point to step 1 again whenever "new requirements" are mentioned.
- Remind students that the system analyst studies the existing system before anything is designed.

ANSWERS

Chapter 2

The system development life cycle

A Label the life cycle

6 marks

1. Identification of requirements	**2.** Designing the solution	**3.** Coding of the solution
4. Testing and debugging	**5.** Deployment of the system	**6.** Maintenance of the system

1 mark each, in this order (page 57). Accept short forms such as "Coding" or "Deployment".

B Which method?

8 marks

1. The developer watches how the books are issued in the library. → **Observation**
2. The developer meets the librarian for a discussion. → **Interviews**
3. The librarian writes answers to a set of questions. → **Questionnaires**
4. The reports and the library cards of the library are examined. → **Document sample collection**
5. Users comment on screenshots of the proposed system. → **Prototyping**
6. The users talk openly about their personal requirements. → **Interviews**
7. At the initial stage, the existing system is observed as it is. → **Observation**
8. The answers collected are analysed to understand the needs. → **Questionnaires**

1 mark each (pages 57 - 58). Accept "an interview", "a questionnaire".

C True or false

6 marks

1. **False** - The first step of the SDLC is designing the solution.
2. **True** - An interview is useful to identify personal requirements.
3. **False** - The tests are planned in the coding step.
4. **False** - Unit testing is done after acceptance testing.
5. **True** - In direct deployment the existing system is totally terminated.
6. **True** - Computers, software and computer networks are all maintained.

D Which level of testing?

8 marks

Which testing?	Unit	Integration	System	Acceptance
Each unit in the system is tested on its own.	☑	☐	☐	☐
Units already tested and debugged are joined and tested together.	☐	☑	☐	☐
The complete system is tested for its outputs for the given inputs.	☐	☐	☑	☐
The user uses the system and approves it or asks for more improvements.	☐	☐	☐	☑
The units of the Accounts branch and the Establishment branch are tested independently.	☑	☐	☐	☐
It is carried out at the late stages of testing, with the users.	☐	☐	☐	☑
The integrated system is tested.	☐	☑	☐	☐
After integration testing, the whole system is tested with given inputs.	☐	☐	☑	☐

1 mark each (Figure 2.5, pages 59 - 60).

E Which deployment approach?

8 marks

Which approach?	Direct	Parallel	Pilot	Phased
The manual library system is abandoned and new library software is installed.	☑	☐	☐	☐
The new library system is used together with the existing manual system.	☐	☑	☐	☐
A system for the whole country is first installed in a selected district.	☐	☐	☑	☐
Dictionaries first, course books second and fiction last.	☐	☐	☐	☑
The existing system is totally terminated and the new system is installed.	☑	☐	☐	☐
A revenue collection system for vehicles is first installed in one province.	☐	☐	☑	☐
The old and the new systems are used together for a certain period.	☐	☑	☐	☐
The new system is deployed in stages: success with each stage leads to the next.	☐	☐	☐	☑

1 mark each. The library, the country, the revenue and the book examples are those of pages 60 - 61.

ANSWERS

Chapter 3

Life cycle models

A Complete the waterfall

4 marks

1. Planning solution	**2.** Implementation of solution	**3.** Testing and debugging of solution	**4.** Deployment of the system

1 mark each (Figure 2.6). The extra names are **Evaluation** and **Initial planning**: they belong to Figure 2.7.

B True or false

5 marks

1. **True** - Each SDLC model facilitates system development in a different way.
2. **False** - In the waterfall model, the next stage can start before a stage is complete.
3. **False** - In the waterfall model the user sees the system at the beginning of the life cycle.
4. **True** - In the iterative incremental model each iteration develops more components.
5. **False** - The iterative incremental model starts with a complete, final list of requirements.

C Waterfall or iterative incremental?

8 marks

The model ...	Waterfall	Iterative incremental
Complete one stage before going to the next phase.	☑	☐
Develop one small component at a time.	☐	☑
The result of the developed system is found at the final stage.	☑	☐
Developers use the knowledge gained in the earlier iterations.	☐	☑
It shows linear development.	☑	☐
The first step begins with a rough idea of the system requirement.	☐	☑
The user sees the system only at the end of the life cycle.	☑	☐
At each stage the system becomes larger, with more components.	☐	☑

1 mark each (pages 61 - 63).

D Name the four models

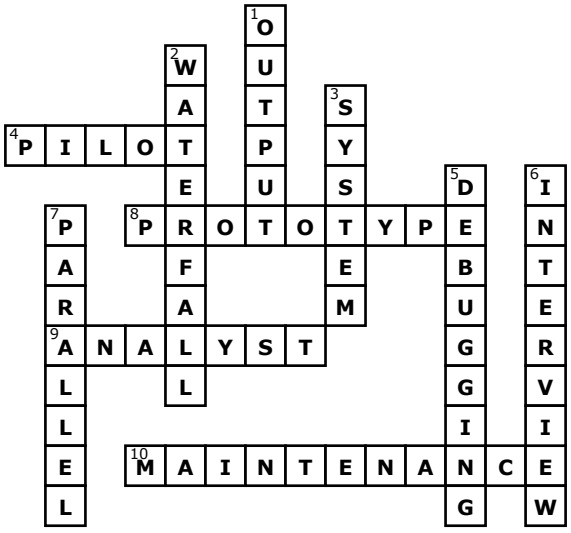
4 marks

1. Waterfall Model	**2.** Iterative Incremental Model	**3.** Prototyping Model	**4.** Spiral Model

1 mark each, in any order (page 61).

E Crossword

10 marks



Across: 4 PILOT, 8 PROTOTYPE, 9 ANALYST, 10 MAINTENANCE

Down: 1 OUTPUT, 2 WATERFALL, 3 SYSTEM, 5 DEBUGGING, 6 INTERVIEW, 7 PARALLEL

★ TEACHING TIPS

- Role-play an interview: one student is the librarian, one is the system analyst with a notebook.
- Use Practical 3 as a pair discussion, then write the best comparison on the board.

MARKING SCHEME

Unit 2 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (c) good citizens	**2.** (b) information
3. (d) manually	**4.** (c) system analyst
5. (a) sees	**6.** (b) e-books for members
7. (d) waterfall	**8.** (b) A and B

2 Fill in the blanks

8 marks (1 each)

1. children	**2.** fines	**3.** index	**4.** report
5. inputs	**6.** bar	**7.** reservation	**8.** system

3 Label and name the pictures

8 marks (1 each)

1. Unit testing	**2.** Integration testing	**3.** System testing	**4.** Acceptance testing
5. Pilot deployment	**6.** Direct deployment	**7.** Phased deployment	**8.** Parallel deployment

Boxes 1 - 4 must be in this order (Figure 2.5). Cards 5 - 8: accept "pilot", "direct", "phased" and "parallel" without the word deployment.

4 Answer these

16 marks

1. What is a system? (2 marks)

A collection of components (1 mark) that interact to achieve a specific task (1 mark).

2. Write two differences between manual systems and computer based information systems. (2 marks)

Any two rows of the table of page 56 (1 mark each): errors (minimal / more room for error); efficiency; storage space (small space, database software / filing cabinets); security (backups and passwords / open to threats).

3. Name the six steps of the System Development Life Cycle in order. (3 marks)

Identification of requirements, designing the solution, coding of the solution, testing and debugging, deployment of the system, maintenance of the system (3 marks for all six in order; 2 for five, or for six in the wrong order; 1 for three or four).

TIP: Ask the class why the waterfall model can give the user a system that he or she did not want (page 62).

4. Name three methods of gathering information for the identification of requirements. (3 marks)

Any three (1 mark each): observation, interviews, questionnaires, document sample collection, prototyping.

5. Why does a system need maintenance? Write two reasons. (2 marks)

Any two (1 mark each): to suit new user requirements; to solve issues that were not identified during development; to adopt new technology to improve efficiency.

6. Write the limitation of the waterfall model. (2 marks)

The user sees the system only at the end of the life cycle (1 mark), so the system developed may be different from what the user wants (1 mark).

7. Write two features of the iterative incremental model. (2 marks)

Any two of page 63 (1 mark each): one small component at a time, each iteration develops more components; uses the knowledge gained in earlier iterations; begins with a rough idea of the requirements; the system grows larger at each incremental stage.

WHAT STUDENTS USUALLY GET WRONG

- They call one part (the computer) the system. A system is the **whole collection** of interacting components.
- They mix up data and information: an information system converts **data into information**.
- They swap the input and the output of the examples: the school takes in **children** and gives **good citizens**.
- They put "planning of tests" under testing. It is an activity of **designing the solution** (step 2).
- They mix up integration testing (the tested units joined) and system testing (the complete system with given inputs).
- They mix up **pilot** (first in a selected area) and **phased** (in stages) deployment.
- They forget the limitation of the waterfall model: the user sees the system only at the end.

TEACHING TIPS

- Explain the testing levels with a bicycle: test each part, join the parts, ride it, and let the owner try it.
- Act out the deployment approaches with the class timetable: change it all at once, both for a week, one class first, or one day at a time.
- Ask: which deployment approach is the safest for the school library? Let students give reasons.
- The waterfall model: water does not flow back up. Each stage is completed before the next one starts.
- Let pairs swap books and check the tick tables before you mark them.

NOTES ON THE TEXTBOOK (PAGES 53 - 63)

- p. 56 "stored in a small physical surface" means a small physical **space**.
- p. 59 "Errors in cording" means errors in **coding**. p. 60 "carried out at with the users" and "improvments" are misprints.
- p. 60 the direct deployment example is garbled: the manual library system is abandoned and new library management software is installed.
- p. 62 Figure 2.7 prints "Evolution": the iterative development diagram says **Evaluation** (the workbook prints Evaluation - accept both). Its unlabelled arrow on the right is left out.
- Figure 2.6 calls steps 2 and 3 "Planning solution" and "Implementation of solution" (page 57: designing and coding).
- p. 63 "componant" = component, "Compere and contrust" = compare and contrast, "life cyle" = life cycle.
- Page 61 names the prototyping and spiral models but does not explain them: only their names are taught.

HOMEWORK AND EXTRA PRACTICE

- Choose a system at home or in your village (a shop, a bus service, a kitchen). Write its input, process and output.
- List four records your school office keeps by hand. Which would be easier in a computer based system? Why?
- Draw the SDLC cycle from memory and write one sentence about each step.
- Make a table of the four deployment approaches with the textbook examples.
- Fast learners: plan the deployment of a computer based library system for your school. Which approach, and why?

TEACHING TIPS

- Build a simple model of the iterative incremental model with four puzzle pieces: one piece in every round.
- Read the summary of textbook page 63 with the class. It matches the revision page of the workbook.
- Give a five-question oral quiz at the start of each lesson. Use the list on the practicals page.
- The writing spaces are revision notes: read each note slowly, or write it on the board, and check the books.
- Mark the unit test together the next day. Students learn most from the questions they got wrong.
- Ask for the six SDLC steps in order at the start of every lesson until everyone knows them by heart.
- Revise with the crossword in pairs: one reads the clue, the other writes the word.

WRITING SPACES

What Students Write

Read each note out or write it on the board - the students copy it into the space

Chapter 1

No.	Page	What the students write
1	4	**What a system is** A system is a collection of components that interact to achieve a specific task. Its three basic components are input, process and output.
2	5	**Manual or computer based?** Computer based: minimal errors, efficient, large data in a small space, secure with backups and passwords. Manual: more errors, less efficient, filing cabinets, open to threats.

Chapter 2

No.	Page	What the students write
3	8	**The six steps of the SDLC** 1 Identification of requirements, 2 Designing the solution, 3 Coding, 4 Testing and debugging, 5 Deployment, 6 Maintenance. New requirements: back to step 1.
4	10	**The four levels of testing** Unit: each unit alone. Integration: the tested units joined. System: the complete system. Acceptance: by the users.
5	10	**The four deployment approaches** Direct: old stopped, new installed. Parallel: both used for a time. Pilot: a selected area first. Phased: in stages.

Chapter 3

No.	Page	What the students write
6	14	**The waterfall model** Linear: identify the requirements first, then complete one stage before the next. The user sees the system only at the end.
7	14	**The iterative incremental model** Begins with a rough idea of the requirements and develops one small component at a time. The system grows with each iteration.

TEACHING TIPS

- Praise answers that use the textbook words: requirements, prototype, integration, pilot, iteration.
- Students who finish early can write two more questions for the oral quiz.
- Use the school canteen as a system example: orders in, cooking and serving, meals out.
- Ask students to bring one newspaper example of a new system that was first tried in one district (pilot).